LAB EXPERIMENT RECORD

20-10-2026---Set 24: Product Inventory Management

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Day	Tuesday

Aim

To analyze product inventory data in R and Tableau using a bar chart, stacked bar chart, scatter plot and an interactive dashboard.

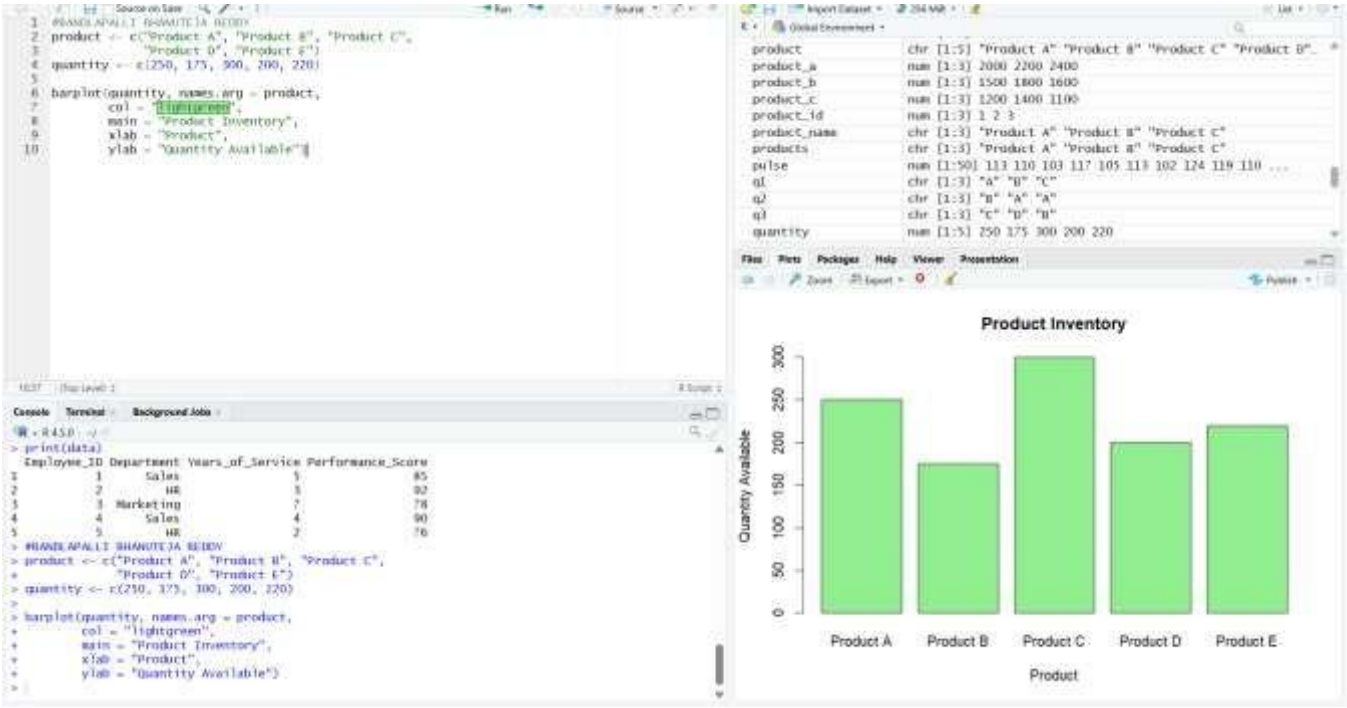
Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Question 1: Bar Chart of Product Quantity

Question: In R, create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

Aim: To visualize the quantity available for each product using a bar chart with labelled axes and a title.

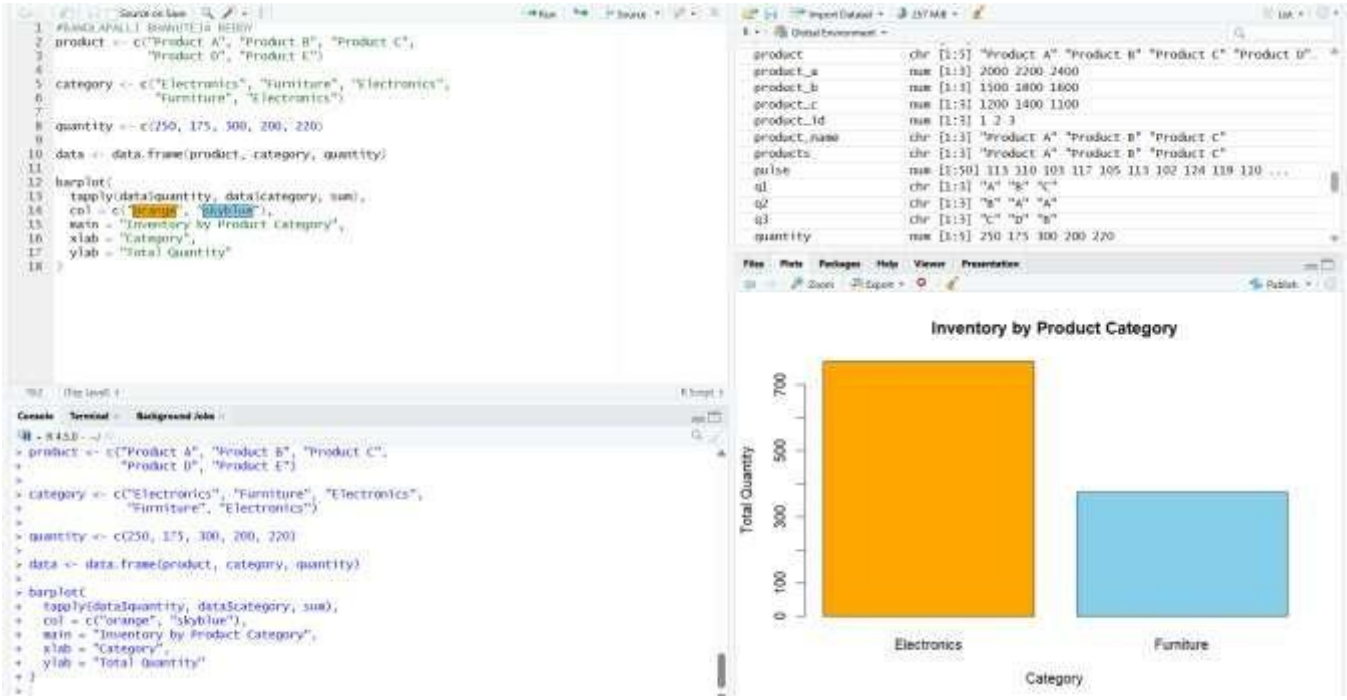


Result: Product C has the highest stock (300) and Product B the lowest (175), so Product B needs restocking first.

Question 2: Stacked Bar Chart by Product Category

Question: In R, generate a stacked bar chart to show the quantity of each product within different product categories.

Aim: To show how the quantity of each product is divided across product categories using a stacked bar chart.

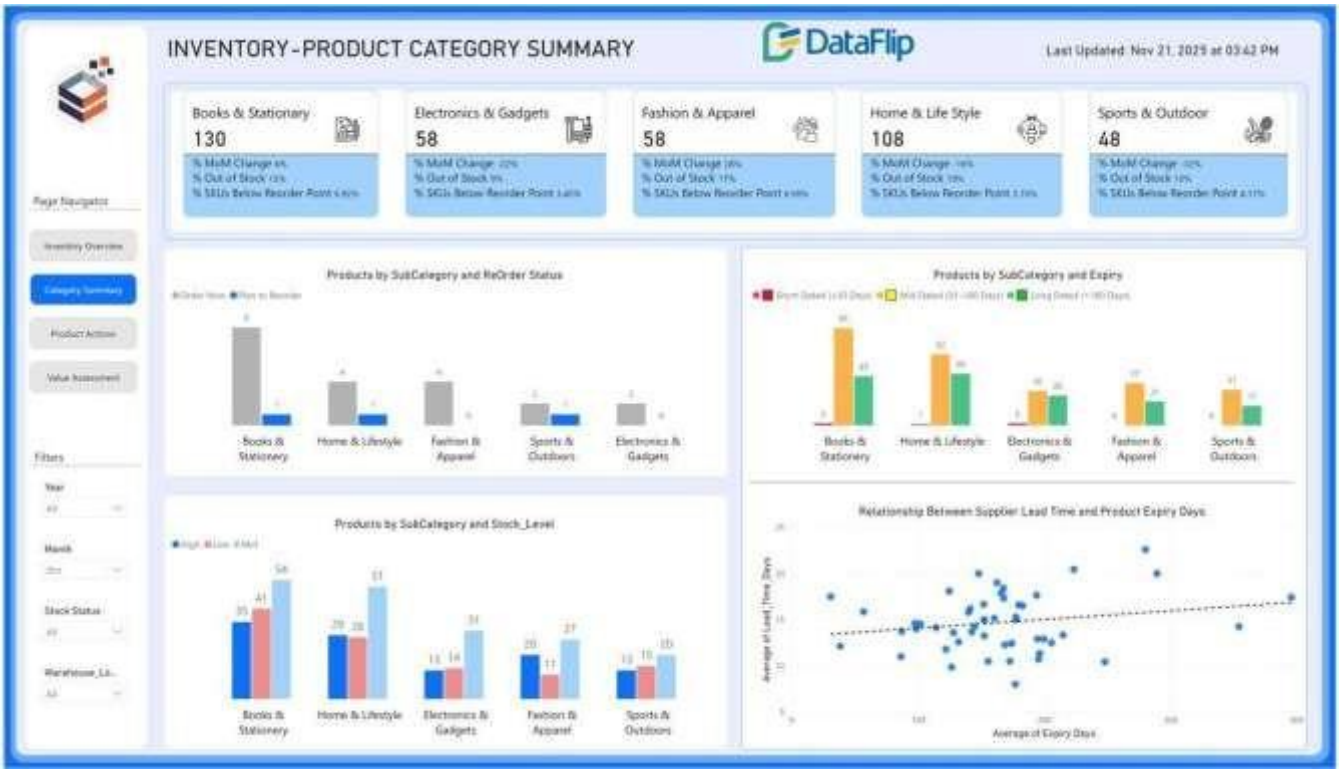


Result: Each bar is split by category, showing each category's share of a product's stock; the dataset gives no category column, so categories must be supplied.

Question 3: Interactive Tableau Dashboard

Question: In Tableau, build a dashboard combining the bar chart and stacked bar chart for interactive exploration of inventory data.

Aim: To build an interactive Tableau dashboard that combines the bar chart and stacked bar chart for exploring inventory data.

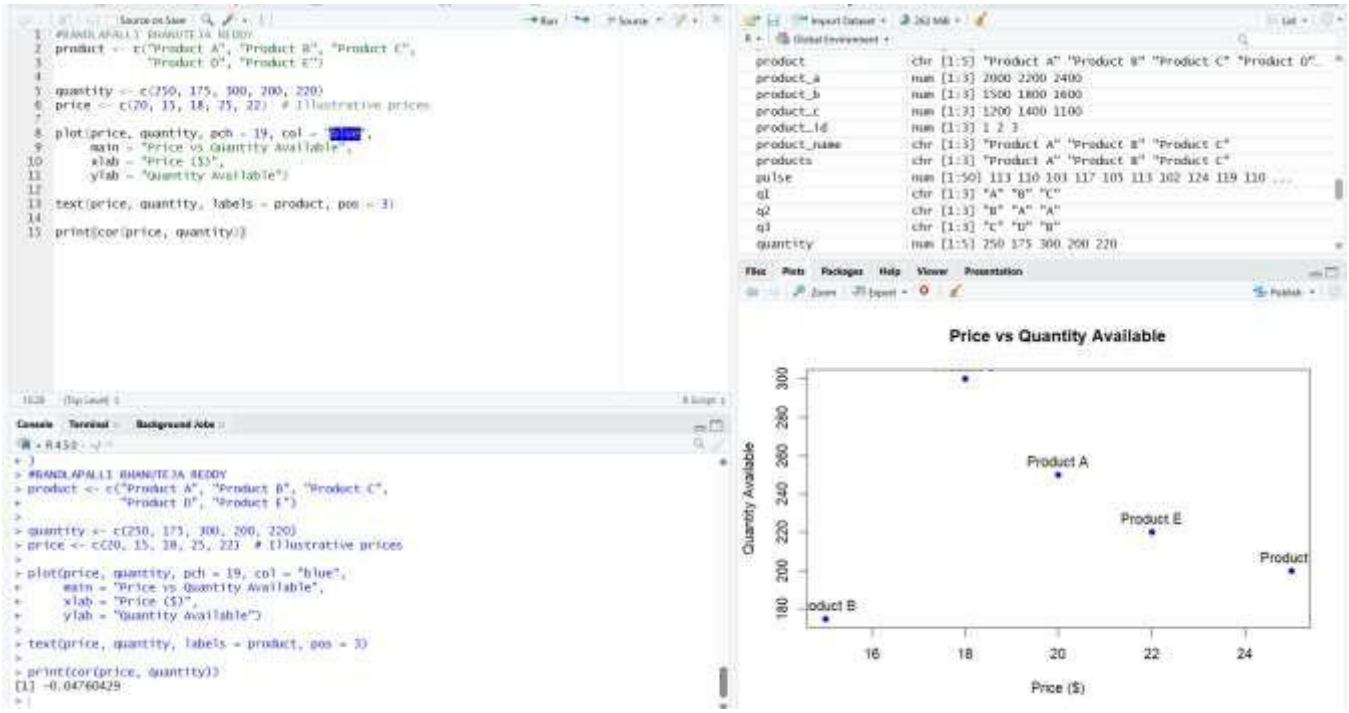


Result: The dashboard places both charts together, and a product filter updates them so users can explore the inventory interactively.

Question 4: Scatter Plot of Price vs Quantity

Question: In R, create a scatter plot to explore the relationship between product price and quantity available. Explain any insights.

Aim: To explore how product price and quantity available are related using a scatter plot.



Result: The plot shows whether higher-priced products are stocked in smaller quantities; the dataset has no price column, so a conclusion needs price values.

Overall Result

Observation: Total stock is 1145 units with an average of 229 per product, and Product C is the largest (300) while Product B is the smallest (175).